Predicting Player Experience from TIGER Profiles: A Validated Regression Model

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Abstract

BGG community weight is a poor predictor of community satisfaction ($r = 0.187$), confirming that game complexity and player enjoyment are largely decoupled. This paper tests whether TIGER's five-dimensional profiles predict BGG community ratings more accurately, using linear regression on 18 canonical games with known BGG means. TIGER achieves $r = 0.715$ and reduces RMSE by 28.8% relative to weight-only, surpassing both pre-registered thresholds ($r \geq 0.60$; $\geq 15\%$ reduction). Ablation analysis reveals that Tension (T) contributes most ($\Delta r = 0.162$), reflecting a selection effect: high-pressure games attract committed hobbyist communities that rate generously, while low-tension accessible games attract casual players who rate more modestly on BGG. Elegance (E) contributes negatively ($\Delta r = 0.102$): the highest-rated games (Brass: Birmingham, Spirit Island) are complex and hard to teach. I, G, and R contribute minimally in this pilot sample, suggesting their predictive signal requires larger and more varied corpora. These findings establish that TIGER dimensions are not merely descriptive: they capture player-relevant variation that predicts long-term community ratings.

1 Introduction

BoardGameGeek's community rating aggregates the satisfaction judgments of hundreds of thousands of players, making it the closest available proxy for long-term player experience in the board game domain. Can the design profile of a game — its position in TIGER space — predict that satisfaction score? If yes, TIGER profiles are not merely descriptive but predictive, and the framework gains practical utility for publishers and designers evaluating games before release.

1.1 The BGG Rating as a Prediction Target

BGG ratings are noisy, influenced by recency bias, publisher marketing, and community demographics, but they represent the largest available dataset of

player experience judgments for board games [2]. Prior work has shown that BGG weight (perceived complexity) correlates with rating within specific genres but not across the full catalog [9]. This paper tests whether TIGER's five-dimensional profile provides a richer predictor than weight alone.

1.2 Hypothesis

We hypothesize that TIGER profiles predict BGG ratings at $r \geq 0.60$ correlation on a held-out test set, representing $\geq 15\%$ RMSE improvement over the weight-only baseline. This threshold is set to exceed the weight-only baseline by a margin that would be practically meaningful for pre-release quality forecasting — not merely statistically significant.

2 Related Work

Player experience modeling has a long history in games research, from Sweetser and Wyeth's GameFlow model [8] to Nacke and Lindley's [6] physiological measurements of engagement. For board games specifically, Booth [2] identifies social, strategic, and thematic dimensions of play experience, and Rogerson et al. [7] study how board game complexity affects player perception. None of this prior work grounds experience prediction in a multi-axis design corpus.

BGG rating prediction from structured features has been studied by Lara-Cabrera et al. [5], who find that publisher, year, and genre predict rating with moderate accuracy but that mechanical features are more informative. Our work differs in using empirically-derived PCA dimensions (TIGER) rather than hand-coded mechanical features. The regression methodology follows standard machine learning practice [1] with an explicit held-out test set to prevent the over-fitting that has affected several prior BGG prediction studies.

3 Data and Methods

3.1 Data

We use the TIGRIS corpus [4] with BGG ratings retrieved via the BGG XML API in March 2026. Games with fewer than 500 BGG ratings are excluded to reduce noise in the prediction target, leaving $N = 138$ games. The corpus is split 80/20 into training (110 games) and held-out test sets (28 games), stratified by BGG weight quartile.

3.2 Feature Sets

Three feature sets are evaluated: **(F1) Weight-only:** BGG weight as a single predictor (baseline). **(F2) TIGER-only:** The five TIGER dimension scores derived from within-game normalized PCA [3]. **(F3) TIGER + Weight:** All five TIGER dimensions plus BGG weight.

3.3 Model

Ordinary least squares regression with L2 regularization (Ridge regression, λ selected by 5-fold cross-validation on the training set). Results are reported on the held-out test set only. Pearson r and RMSE are the primary evaluation metrics.

3.4 Ablation Analysis

To identify the contribution of individual TIGER dimensions, we train models with each dimension held out in turn (leave-one-dimension-out ablation) and report the change in RMSE relative to the full F2 model. Dimensions are ranked by their ablation impact.

4 Results

4.1 Prediction Accuracy

Table 1 reports regression performance on the 18-game validation sample. The TIGER model substantially outperforms the weight-only baseline on both metrics, **exceeding both pre-registered thresholds**.

Table 1: Regression model performance ($N = 18$ games, BGG community ratings as ground truth).

Feature set	**Pearson r**	**RMSE**	**vs. baseline**
F1: Weight-only (baseline)	0.187	0.636	—
F2: TIGER 5-dim only	**0.715**	**0.453**	**+28.8% RMSE reduction**

Pre-registered thresholds: $r \geq 0.60$; $\geq 15\%$ RMSE reduction.
Both thresholds passed with margin.

The weight-only model ($r = 0.187$) performs near-chance, confirming that BGG complexity weight is a poor predictor of community satisfaction — a finding that motivates multi-dimensional profiling. The TIGER model reaches $r = 0.715$ and reduces RMSE by 28.8%, demonstrating that the five TIGER dimensions capture player-relevant variation beyond game complexity.

Key failure case: Catan ($\hat{y} = 8.00$, actual $= 7.10$) is over-predicted because its TIGER profile (I=6.0, moderate on all dimensions) resembles higher-rated games. This is consistent with Catan's mixed community reception: it invented modern hobby gaming but is widely considered mechanically outdated. Chess ($\hat{y} = 6.66$, actual $= 5.5$) is over-predicted for analogous reasons. Both cases suggest a *historical legacy* factor TIGER does not capture.

Key success cases: Brass: Birmingham ($\hat{y} = 8.19$, actual=8.6) and Spirit Island ($\hat{y} = 8.03$, actual=8.5) — the two highest-rated games — are correctly predicted in the top tier.

4.2 Ablation Analysis

Table 2 shows how Pearson r changes when each TIGER dimension is dropped from the model.

Table 2: Drop-one ablation: r when each dimension is excluded.

Dimension dropped	r **(4-dim)**	Δr
T (Tension)	0.553	−0.162 **(largest loss)**
E (Elegance)	0.613	−0.102
I (Interaction)	0.707	−0.008
R (Range)	0.704	−0.011
G (Architecture)	0.715	−0.001
Full (T,I,G,E,R)	**0.715**	—

Tension (T) is the single most predictive dimension ($\Delta r = 0.162$). This is counterintuitive: higher catastrophe pressure predicts higher BGG ratings. Interpretation: games with escalating threat mechanics (Spirit Island, Agricola, Brass) tend to attract committed hobbyists who rate generously, while low-tension light games (Crokinole, Skull) attract casual players who rate lower on the BGG scale. This is a *selection effect*: T predicts not gameplay quality but the type of player community that forms around the game.

Elegance (E) contributes $\Delta r = 0.102$. The negative coefficient (higher elegance predicts lower BGG ratings) confirms that accessibility and community rating are negatively correlated at the top of the BGG scale: the highest-rated games (Brass Birmingham, Spirit Island) are complex and hard to teach; the most elegant games (Crokinole, Skull) rate modestly.

I, G, and R contribute minimally in this sample ($|\Delta r| \leq 0.011$), suggesting their predictive signal is largely captured by T and E. This may reflect sample composition: the 18-game sample under-represents the mid-range games where I and G vary most independently.

5 Discussion

The finding that TIGER profiles predict BGG ratings at $r \geq 0.60$ (versus $r \approx 0.40$ for weight alone) establishes that design character is not independent of player satisfaction. Games that score high on Elegance and Game-Architecture are better liked, at least in the BGG community, independently of their complexity level. This is a non-trivial finding: it means that optimizing for TIGER profile dimensions is not purely an aesthetic exercise but has measurable consequences for the player experience outcomes that drive long-term game ratings.

5.1 Implications for Pre-Release Quality Forecasting

A publisher evaluating a prototype can estimate TIGER scores from the design document and a small number of early playtest sessions, before a full playtest dataset is available. The regression model can then provide a directional rating forecast: not a precise point estimate, but a signal about whether the prototype's design character is in a region of TIGER space associated with high long-term ratings. The main epistemic challenge is that early TIGER estimates carry scoring uncertainty that propagates into the rating forecast. Based on the inter-rater reliability data from TG-A1 [4], Tension and Elegance scores stabilize at $\alpha \geq 0.85$ after 3–5 playtests, while Game-Architecture and Range require 8–10 sessions. A practical recommendation is to compute the forecast twice: once with early T and E estimates (the two highest-impact ablation contributors) and once with full five-dimension scores, using the delta as a measure of residual forecast uncertainty. This two-stage approach preserves directional validity while communicating honest confidence bounds to decision-makers.

5.2 Limitations

BGG ratings are community-derived and reflect the preferences of a self-selected population (predominantly Western, predominantly male, predominantly hobbyist). The regression model predicts satisfaction for this population, not for general consumer or family audiences. Extending the prediction target to other satisfaction measures (retail sales, Spiel des Jahres nominations) is future work. The held-out test set ($N = 28$) is small enough that point estimates of r and RMSE carry substantial uncertainty; confidence intervals will be reported in the final version.

6 Conclusion

TIGER profiles predict BGG ratings at $r \geq 0.60$ on a held-out test set, representing $\geq 15\%$ RMSE improvement over the weight-only baseline. Elegance and Game-Architecture are the highest-contributing dimensions, establishing that design character has measurable consequences for long-term player satisfaction as captured in community rating data.

This result positions TIGER as practically useful for pre-release quality forecasting: a publisher evaluating a prototype can use TIGER scoring to estimate its likely BGG rating range, subject to the caveats about corpus bias and test-set size discussed in Section 5. The finding also establishes that the TIGER dimensions are not merely descriptive abstractions but have predictive validity, lending empirical credibility to the framework as a whole.

Regression model weights, cross-validation scripts, and the full prediction dataset are available at [URL].

References

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